

FITBUDDY

Requirement Analysis Phase

Project: FitBuddy – A Personal Fitness and Wellness Management System

1. Introduction

FitBuddy is a fitness and wellness application designed to help users manage their daily fitness activities, workouts, diet, and progress in one place. The system provides personalized fitness support and helps users maintain healthy habits.

2. Objectives

- Provide personalized workout plans based on user needs.
- Help users track daily workouts and fitness activities.
- Allow users to record and monitor diet and water intake.
- Track fitness progress such as weight, calories, and workout duration.
- Send reminders and notifications for workouts and healthy habits.
- Provide a simple and user-friendly interface.

3. Stakeholders

Stakeholder	Role
User	Uses FitBuddy to manage fitness and wellness activities.
Fitness Trainer	Provides workout guidance and monitors user progress.
Administrator	Manages users, content, plans, and system settings.

4. Functional Requirements

- **User Registration and Login:** Users can create an account and securely log in.
- **User Profile:** Users can enter age, gender, height, weight, fitness goal, and activity level.
- **Workout Management:** The system provides workout routines and exercise details.
- **Diet Tracking:** Users can record meals, calories, and water intake.
- **Progress Tracking:** Users can view changes in weight, calories, workout time, and other fitness records.
- **Notifications:** The system can remind users about workouts, water intake, and other activities.
- **Trainer Support:** Trainers can provide plans and guidance where applicable.
- **Admin Management:** Administrators can manage users, workout content, and system data.

5. Non-Functional Requirements

- **Usability:** The application should be simple and easy to use.
- **Performance:** Pages and user actions should respond quickly.

- **Security:** User information and login credentials must be protected.
- **Reliability:** User fitness records should be stored accurately.
- **Scalability:** The system should support increasing numbers of users.
- **Availability:** The application should be available whenever users need it.
- **Maintainability:** The software should be easy to update and maintain.

6. Hardware and Software Requirements

Category	Requirement
Device	Smartphone, tablet, or computer
Operating System	Android / iOS / Windows or web-compatible OS
Internet	Internet connection for online features
Database	A suitable database such as MySQL or Firebase
Development	Web or mobile development environment

7. User Requirements

- Easy registration and login.
- Simple dashboard showing daily fitness information.
- Personalized workout and diet information.
- Easy tracking of daily activities.
- Clear progress reports.
- Reminders for regular fitness activities.

8. Expected Outcome

The requirement analysis phase identifies the main users, system functions, technical needs, and quality requirements of FitBuddy. These requirements will serve as the foundation for the design and development of the FitBuddy system.